

IEOR 4404 – Fall 2025
Solutions for Practice Midterm (Structural Variant B)

Problem 1 (Zero-truncated Poisson). Let X be $\text{Poisson}(\lambda)$ conditioned on $\{X \geq 1\}$, i.e. $\mathbb{P}(X = i) = \frac{e^{-\lambda} \lambda^i / i!}{1 - e^{-\lambda}}$ for $i \geq 1$.

(a) **Inverse transform via recursion.** Let $p_i = \mathbb{P}(X = i)$. Then

$$p_1 = \frac{e^{-\lambda} \lambda}{1 - e^{-\lambda}}, \quad \frac{p_{i+1}}{p_i} = \frac{\lambda}{i+1}, \quad i \geq 1.$$

Algorithm.

1. Generate $U \sim \text{Unif}(0, 1)$. Initialize $i = 1$, $q = p_1$, $F = q$.
2. While $U > F$: set $i \leftarrow i + 1$, update $q \leftarrow q \cdot \frac{\lambda}{i}$, and $F \leftarrow F + q$.
3. Return $X = i$.

(b) **Acceptance–rejection interpretation.** Proposal: $Y \sim \text{Pois}(\lambda)$ with pmf $g(k)$. Target:

$$f(k) = \begin{cases} \frac{g(k)}{1 - e^{-\lambda}}, & k \geq 1, \\ 0, & k = 0. \end{cases}$$

Then $C = \sup_k \frac{f(k)}{g(k)} = \frac{1}{1 - e^{-\lambda}}$, the A/R test accepts iff $k \geq 1$, and the acceptance probability is $1/C = 1 - e^{-\lambda}$.

Problem 2 (Lomax / Pareto II). Let $F(x) = 1 - (1 + \theta x)^{-\alpha}$ for $x > 0$, $\alpha, \theta > 0$.

(a) **ITM.** Set $U = F(X)$. Then $1 - U = (1 + \theta X)^{-\alpha}$, so

$$\boxed{X = \frac{(1 - U)^{-1/\alpha} - 1}{\theta}}, \quad U \sim \text{Unif}(0, 1).$$

(b) **A/R with same family (parameter $\tilde{\alpha}$).** Densities: $f(x) = \alpha\theta(1 + \theta x)^{-(\alpha+1)}$, $g(x) = \tilde{\alpha}\theta(1 + \theta x)^{-(\tilde{\alpha}+1)}$. Hence

$$R(x) = \frac{f(x)}{g(x)} = \frac{\alpha}{\tilde{\alpha}} (1 + \theta x)^{\tilde{\alpha} - \alpha}.$$

If $\tilde{\alpha} > \alpha$, then $R(x) \rightarrow \infty$ as $x \rightarrow \infty$ (*lighter-tailed proposal*), so A/R fails. If $\tilde{\alpha} \leq \alpha$, then $R(x) \leq R(0) = \alpha/\tilde{\alpha}$, hence $C = \alpha/\tilde{\alpha}$ is valid and the acceptance rate is $1/C = \tilde{\alpha}/\alpha$.

Problem 3 (Uniform in a piecewise region). Let $g_1(x) = x$ on $0 < x \leq 1$ and $g_2(x) = x^{-3}$ on $x \geq 1$. Area: $\int_0^1 x \, dx + \int_1^\infty x^{-3} \, dx = \frac{1}{2} + \frac{1}{2} = 1$. Thus

$$F_X(x) = \begin{cases} \int_0^x t \, dt = \frac{x^2}{2}, & 0 \leq x \leq 1, \\ \frac{1}{2} + \int_1^x t^{-3} \, dt = 1 - \frac{1}{2x^2}, & x \geq 1. \end{cases}$$

Inverse cdf:

$$F_X^{-1}(u) = \begin{cases} \sqrt{2u}, & 0 \leq u \leq \frac{1}{2}, \\ \frac{1}{\sqrt{2(1-u)}}, & \frac{1}{2} < u < 1. \end{cases}$$

Sampler. Generate $U_1 \sim \text{Unif}(0, 1)$ and set $X = F_X^{-1}(U_1)$. Then generate $U_2 \sim \text{Unif}(0, 1)$ and set $Y = U_2 g(X)$ where $g(X) = X$ if $X \leq 1$ and $g(X) = X^{-3}$ if $X \geq 1$.

Problem 4 (Composition / mixture). Given

$$f_X(x) = \int_0^\infty \frac{4t}{(1+xt)^5} e^{-t} dt, \quad x \geq 0,$$

take $Z \sim \text{Exp}(1)$ and $f_{X|Z=z}(x) = \frac{4z}{(1+xz)^5}$, normalized because $\int_0^\infty \frac{4z}{(1+xz)^5} dx = \int_1^\infty \frac{4}{u^5} du = 1$ by $u = 1+xz$. Thus $F_{X|Z=z}(x) = 1 - (1+xz)^{-4}$ and

$$\boxed{X = \frac{(1-V)^{-1/4} - 1}{Z}}, \quad Z \sim \text{Exp}(1), \quad V \sim \text{Unif}(0, 1).$$